

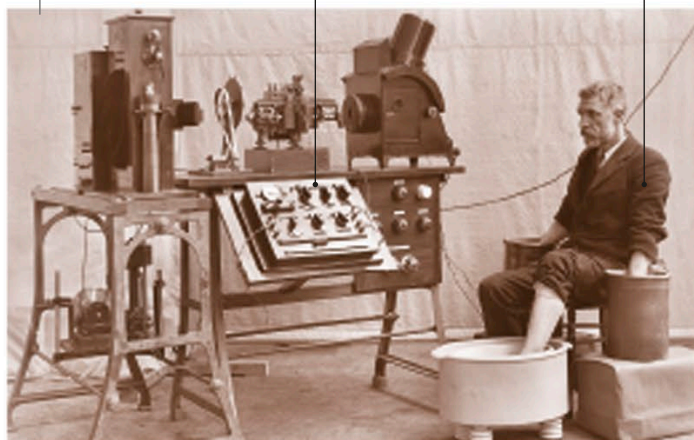
1903

Electrocardiograph

Dutch doctor Willem Einthoven invented the first accurate electrocardiograph (ECG). This machine measures the tiny electric currents generated by the heart as it beats. ECGs are now widely used to detect heart problems.

Machine reads electric signals from the patient conducted via salt water.

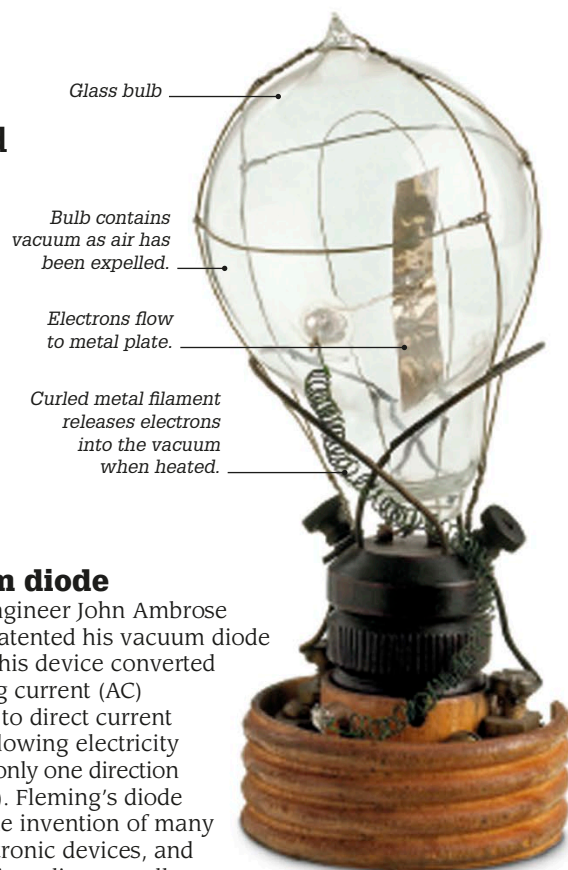
Patient's hands and foot are dipped in salt water.



Patient being scanned by an ECG, 1911



In 1904, Italian businessman Piero Ginori Conti harnessed heat from within Earth, turning water into steam to power the first geothermal electricity generator.



1904

Vacuum diode

English engineer John Ambrose Fleming patented his vacuum diode in 1904. This device converted alternating current (AC) electricity to direct current (DC) by allowing electricity to flow in only one direction (see p.140). Fleming's diode spurred the invention of many early electronic devices, and was used in radios as well as the first computers.

Model of Fleming's diode

1905



THE WRIGHT BROTHERS

American brothers Orville (1871–1948) and Wilbur Wright (1867–1912) ran a bicycle-making business in Dayton, Ohio. Fascinated by flight, the inventors built their own kites, gliders, and a wind tunnel to understand the forces involved in flight. They flew the first heavier-than-air, powered aircraft, the 1903 Wright Flyer (see p.162).



Orville and Wilbur Wright

Wright gliders

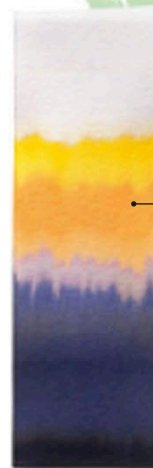
The brothers made more than 200 flights in self-built gliders, learning how aircraft could move in one of three axes: pitch (up and down), yaw (side to side), and roll (see p.163). They developed control surfaces such as a steerable rudder to control yaw and wing warping, with wires bending the wing tips to allow the plane to roll while flying.



Wilbur Wright glides over Big Kill Devil Hill, North Carolina

Chromatography

In 1903, Russian botanist Mikhail Tsvet developed chromatography, a process that can separate out mixtures of pigments, such as in plants or inks. Today, it has many uses, including in forensics.



Different chemical substances travel up the blotting paper at different speeds, separating from each other.

Blotting paper dipped in ink